

Review: Linear Correlation

2 Numerical variables.

Presentation Outline

- 1 Definition
- 2 In Practice
- 3 Properties and Interpretation
- 4 Connection to Causality
- 5 Important points

Linear Correlation between Two Variables

- r_{XY} (or just r for short) is the **sample linear correlation coefficient**.
- r_{XY} measures the **strength** and **direction** of **linear association** between two quantitative variables X and Y .
- To calculate it, use:

$$r_{XY} = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{(n-1)s_X s_Y}$$

where:

- n : number of pairs of observations
- $\bar{X}$, $\bar{Y}$: sample means of X and Y
- s_X , s_Y : sample standard deviations of X and Y

Example: Sibling Heights

Consider 11 families, each with one adult brother and one adult sister. Let X_i be the brother's height (in inches), and Y_i be the sister's height.

i	X_i	Y_i	$X_i - \bar{X}$	$Y_i - \bar{Y}$	$(X_i - \bar{X})(Y_i - \bar{Y})$	$(X_i - \bar{X})^2$	$(Y_i - \bar{Y})^2$
1	71	69	2	5	10	4	25
2	68	64	1	0	0	1	0
3	66	65	-3	1	-3	9	1
4	67	63	-2	-1	2	4	1
5	70	65	1	1	1	1	1
6	71	62	2	-2	-4	4	4
7	70	65	1	1	1	1	1
8	73	64	4	0	0	16	0
9	72	66	3	2	6	9	4
10	65	59	-4	-5	20	16	25
11	66	62	-3	-2	6	9	4
	759	704			39	74	66

$$\bar{X} = \frac{759}{11} = 69,$$

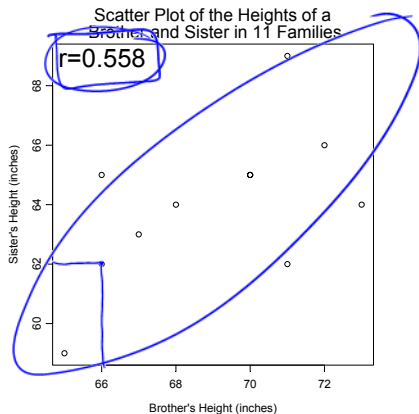
$$\bar{Y} = \frac{704}{11} = 64$$

$$s_X = \sqrt{\frac{74}{10}},$$

$$s_Y = \sqrt{\frac{66}{10}}$$

$$r_{XY} = \frac{3.9}{\sqrt{7.4 \cdot 6.6}} \approx 0.558$$

Scatterplot of Heights



NOTE

- POSITIVE RELATIONSHIP BETWEEN x & y
- SOME INDICATION OF LINEARITY
- NOT VERY STRONG

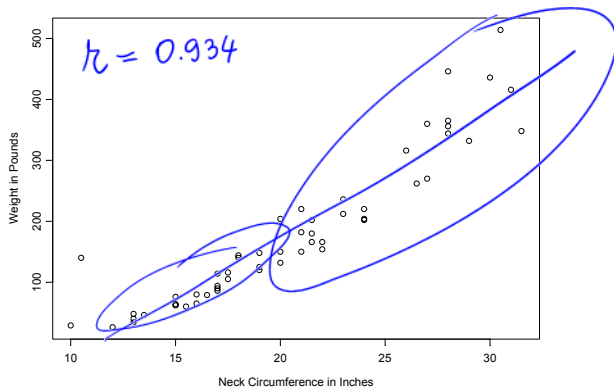
Properties of the Linear Correlation Coefficient

- r is unitless and always lies between -1 and 1 .
- $r = 1$ if and only if all points fall exactly on a line with positive slope.
- $r = -1$ if and only if all points fall exactly on a line with negative slope.
- $r = 0$ implies no *linear* association between X and Y .

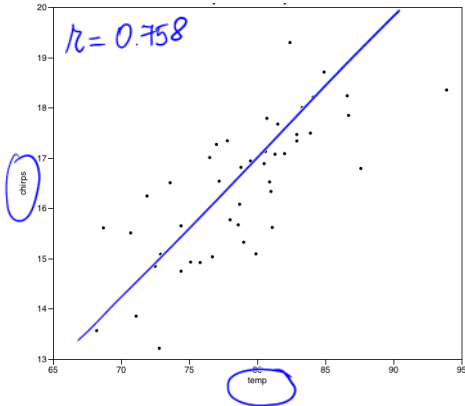


Illustration: No vs Strong Linear Association

Is a Bear's Neck Circumference Correlated with a Bear's Weight?



Do Crickets Chirp More in Warm Weather?

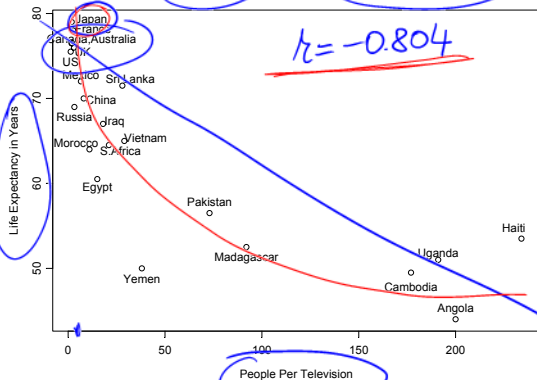


TV Watching and ~~Test Scores?~~

LIFE EXPECTANCY

Does Watching TV Increase Life Expectancy?

$$r = -0.804$$



Beware of Causal Conclusions Based on Correlation

Correlation does not imply causation. Consider the following variable pairs:

Pair	X	Y	Expected r_{XY}
<u>A</u>	Corn stalk diameter	Corn plant weight	High, $r > 0$
<u>B</u>	Person's age	Year of birth	Very negative, $r \approx -1$
<u>C</u>	Daily Dow Jones Avg	Rainfall in Seattle	Near zero, $r \approx 0$
<u>D</u>	# of ultrasounds	Baby's birth weight	Weak or confounded
<u>E</u>	Ice cream sales	Drownings	High, spurious correlation
<u>F</u>	Wife's age	Husband's age	High, $r > 0$



Key takeaways

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- The correlation coefficient r quantifies the strength and direction of a linear relationship between two quantitative variables.
- r is unitless and lies between -1 and 1 .
- A positive r means that as one variable increases, so does the other; a negative r implies an inverse relationship.
- $r = 0$ does not mean no relationship — just no ****linear**** relationship.
- Correlation does ****not**** imply causation! Always consider lurking variables and confounding.
- Use scatterplots alongside r to understand the full context of the relationship.